Explanation

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Experiment : Verification of Nortan's Theorem

Objective

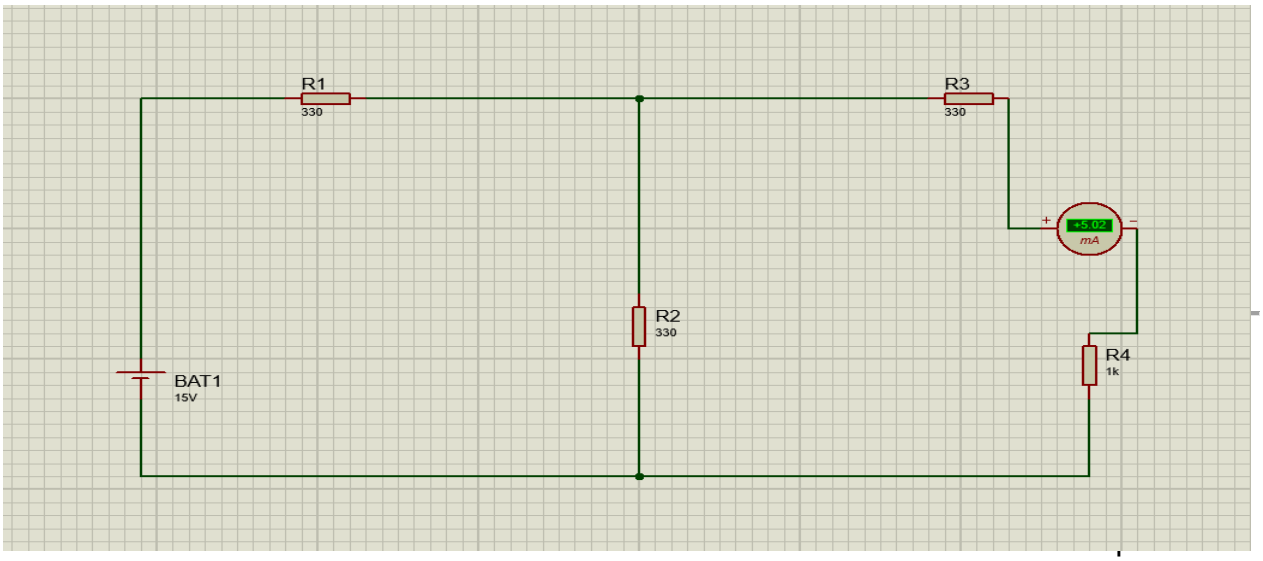
To verify **Nortan's Thorem** by measuring full load current.

Procedure (Steps)

1. Measurement of Original Load Current (IL)

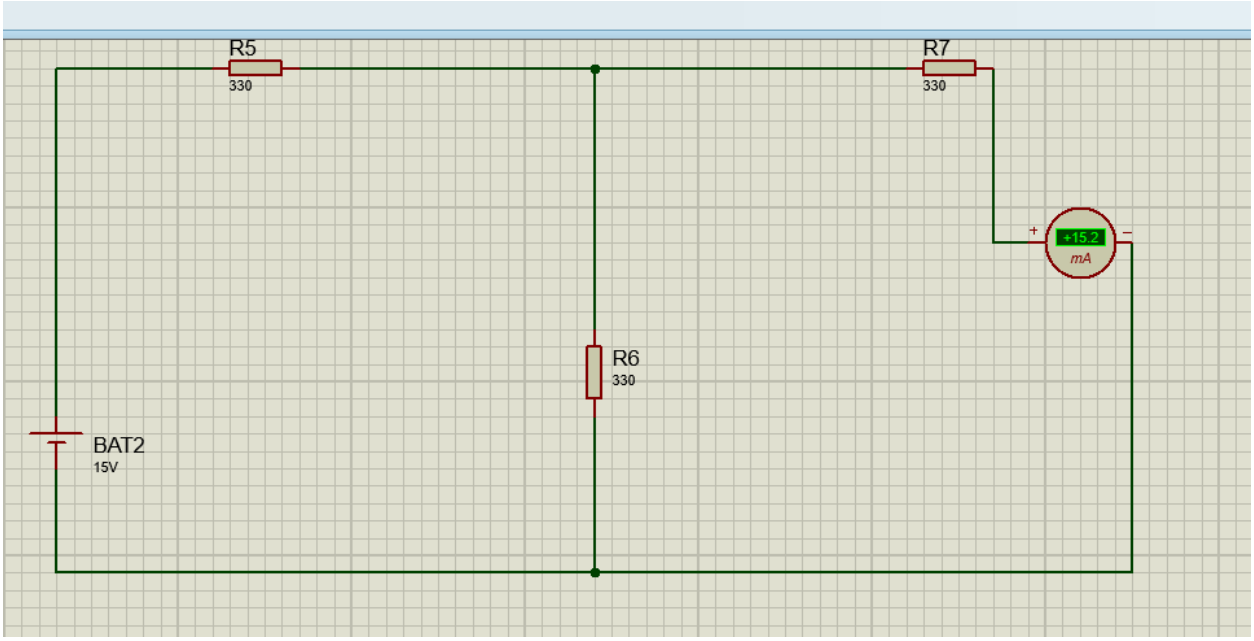
Objective: To find the actual current flowing through the load resistor in the complex circuit.

- **Procedure:** A DC voltage source was set to **15V**. Three **330ohm** resistors and one **1K** load resistor RL were connected in a bridge-T configuration. A **DC Ammeter** (set to mA range) was placed in series with the 1Kresistor.
- **Observation:** The ammeter measured the current IL, which serves as our reference value to verify Norton's Theorem



terminals (where the load was) were "short-circuited" by connecting a **DC Ammeter** directly across them.

- **Observation:** The current recorded on the ammeter is the **Norton Current IN**. This represents the maximum current the circuit can provide when the load resistance is zero.



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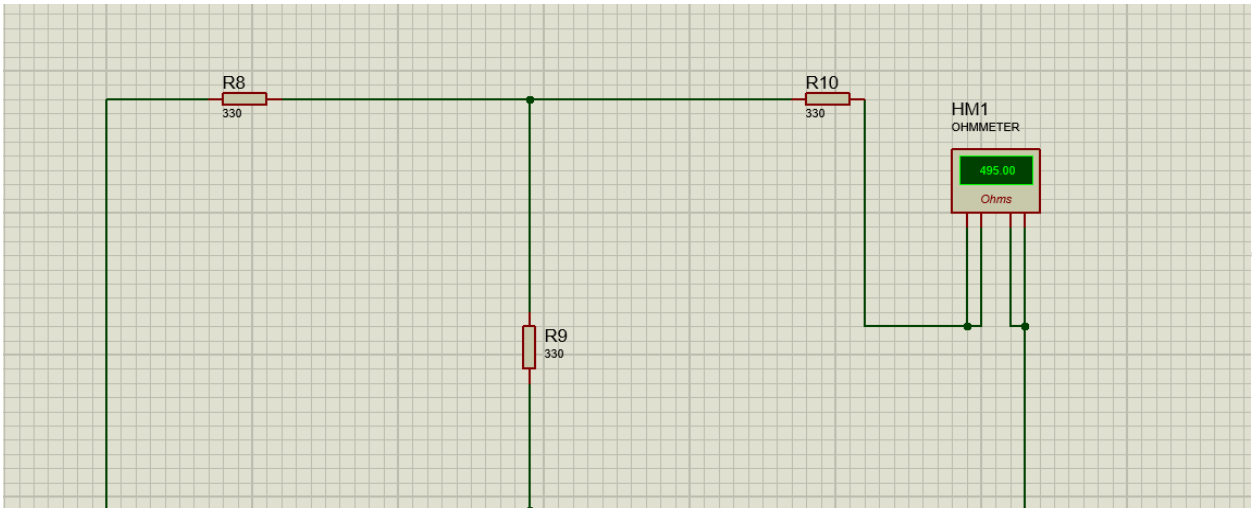
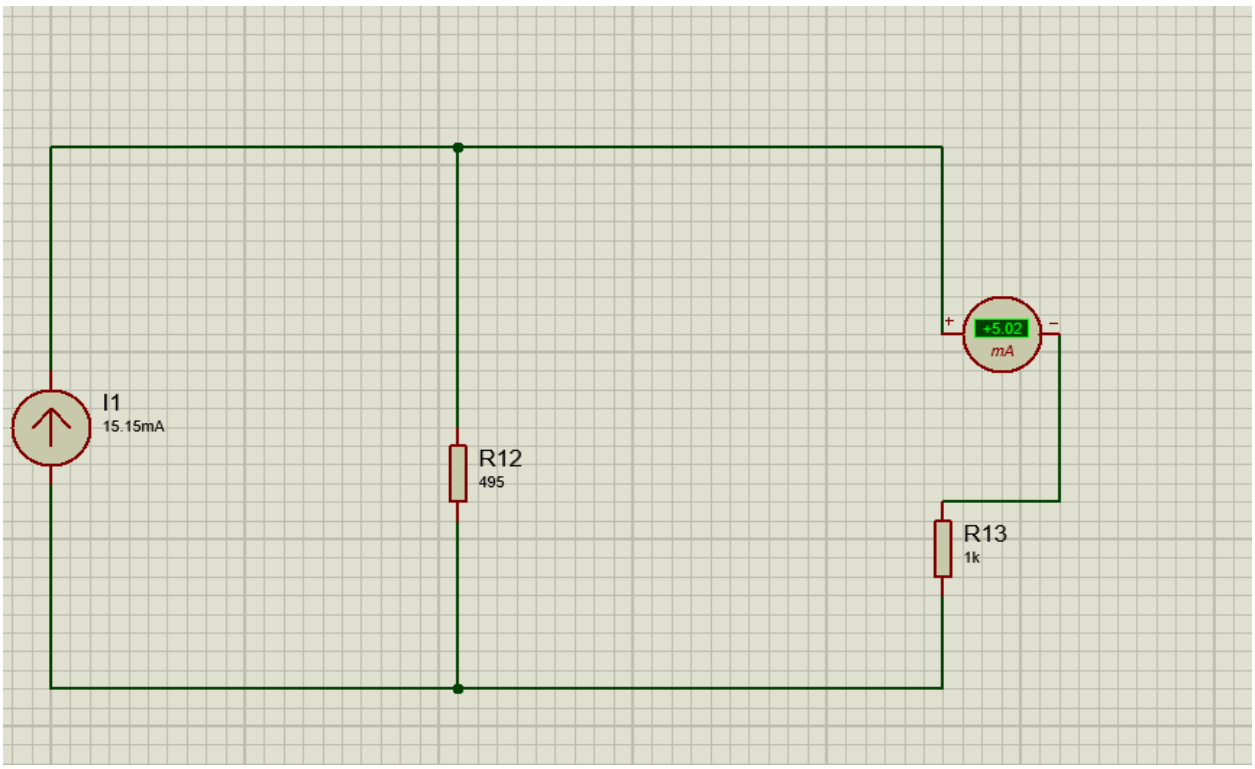
3. Determination of Norton Resistance (RN)

Objective: To find the equivalent internal resistance of the network.

- **Procedure:** The **15V source** was replaced with a short circuit (its internal resistance). The load remained disconnected. An **Ohmmeter** was connected across the open load terminals.
- **Calculation:** Theoretically, we verified this by calculating the parallel combination of the first two 330 ohm resistors and adding the third:



4. Norton Equivalent Circuit Verification

Objective: To prove that a single current source and one parallel resistor produce the same IL as the original complex circuit.

- **Procedure:** A **CSOURCE** (Current Source) was placed and set to the IN value found in step 2 (**15.15mA**). A single resistor representing RN(**495 ohm**) was connected in **parallel** with the current source. Finally, the original **1K ohm** load was connected in parallel to this combination.
- **Observation:** The ammeter in series with the 1k load showed a current of **5.02mA**, which matched the IL from the original circuit, thereby verifying Norton's Theorem